

SU(2), SU(3) and Lie-Algebraic Gauge Structure in Quantum Cosmology

*From catalogue-level anisotropy controls to non-Abelian inflation,
perturbative stability and reheating*

Technical background paper and research status report

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EVIDENCE STATUS

Current claim boundary

The calculations establish reproducible SU(3) embedding, basin and stability boundaries and a survey-sensitive SU(2)/Quaia follow-up candidate.

They do not establish a cosmological SU(2) or SU(3) detection.

Physical promotion still requires a uniquely specified action, fully qualified scalar-vector-tensor propagation, coherent survey nulls and an observational likelihood that defeats foreground and catalogue-systematics explanations.

Abstract

This paper summarises a recent research programme on SU(2) and SU(3) gauge structure in cosmology, written for an audience familiar with general relativity, quantum field theory or observational cosmology.

The programme began with phenomenological SU(2)/SU(2)_R background curves and an angular search in the Gaia-unWISE Quasar catalogue, then moved to an action-level non-Abelian inflation framework in which a pseudoscalar field couples to a homogeneous gauge condensate through a Chern-Simons interaction.

The resulting chain now covers Lie-algebra validation, regular and principal SU(2) embeddings in SU(3), non-perturbative basins of attraction, homogeneous backreaction, adaptive phase boundaries, finite-wavenumber tensor, vector and fully constrained scalar perturbations, dynamical exit surfaces, continuously orthonormalised transfer matrices and microscopic reheating portals.

The principal structural result is an identifiability theorem confirmed numerically: for a constant SU(2) embedding in the minimal Yang-Mills plus pseudoscalar action, homogeneous SU(3) observables and the linearly metric-coupled triplet are equivalent to SU(2) after an embedding-index rescaling of the gauge coupling.

SU(3)-specific information therefore resides in the extra doublet, singlet or quintet representations, in transitions between inequivalent embeddings, in nonlinear or loop effects, or in separately specified hidden-sector dynamics.

The numerical work finds a replicated regular-embedding basin and sharply resolved topology-loss and extinction curves, but the most physically qualified low- ξ K25 background fails the finite- k scalar and positive-helicity vector promotion gates.

The scalar continuation to $x=10^{-4}$ reveals slowly relaxing destructive interference between two axion-dominated modes rather than a ghost. Reheating transfer can be made numerically consistent, but the result remains materially sensitive to portal operators not selected by the parent action.

On the observational side, a completed 300-mock coherent nuisance analysis finds a matched $z=1.0-1.5$, $|b|>10$ deg signal-to-noise ratio of 4.42293.

Its targeted-family probability is 0.00332226, but the all-window probability is 0.0398671; the corresponding priority probabilities are 0.00332226 and 0.0166113.

The local family is therefore interesting, but the preregistered global promotion gate fails.

Earlier catalogue-quality and spatial-block tests also materially altered amplitude, direction or significance.

The correct conclusion is a sharply defined follow-up programme rather than evidence for new physics.

The paper closes with a staged data plan using Quasar, Gaia DR3/DR4, Planck PR4, ACT DR6, Euclid, Rubin, Roman, LiteBIRD and LISA, with explicit promotion criteria for CMB parity, chiral gravitational waves, lensing, large-scale structure and cross-catalogue anisotropy.

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1. Scope, Terminology and Evidence Levels

SU(2) and SU(3) appear in several logically distinct roles.

In particle physics they are internal gauge groups; in cosmology they can support homogeneous non-Abelian fields, source chiral tensor modes, alter thermal histories or describe hidden sectors.

An angular dipole found in a sky catalogue is not itself an SU(2) generator, and an SU(3) background curve is not evidence for quantum chromodynamics.

The research programme therefore separates group theory, action-level dynamics, numerical qualification and observational inference.

Table 1. Evidence ladder used throughout this paper.

Evidence level	Question	What a pass establishes	What it does not establish
L0 Algebra	Are generators, commutators, embeddings and constraints implemented correctly?	Internal mathematical consistency	A physical cosmology
L1 Background	Does a homogeneous solution exist and attract a finite basin?	A viable classical trajectory within the frozen action	Stable perturbations or observability
L2 Perturbations	Are scalar, vector and tensor sectors ghost-free, gradient-stable and controlled?	Linear theoretical qualification	Correct reheating or a data preference
L3 Transfer	Can primordial modes be propagated through exit, reheating and radiation domination?	A complete observable adapter	A detection
L4 Likelihood	Does the model beat Λ CDM and survey templates in held-out data?	Empirical support under explicit priors	Unique new physics without rival-model checks

Notation warning

The project contains a late-time phenomenological SU2/SU2R fluid, a published high-temperature SU(2)-CMB benchmark, and an axion-gauge SU(2)/SU(3) inflation branch.

They are not assumed to be the same physical theory.

A common label is not an action-level identity.

2. Lie Algebra Foundations

2.1 Continuous groups and local generators

A Lie group is a smooth manifold whose elements also form a group.

Near the identity, finite transformations are generated by a Lie algebra.

For a unitary representation with real coordinates θ^a and Hermitian generators T^a , the local structure is:

Equations 1-4. Group element, Lie bracket, generator normalisation and Jacobi identity.

$$\begin{aligned}
 U(\theta) &= \exp(i \theta^a T^a) & (1) \\
 [T^a, T^b] &= i f^{abc} T^c & (2) \\
 \text{Tr}(T^a T^b) &= \delta^{ab}/2 & (3) \\
 f^{abe} f^{cde} + f^{bce} f^{ade} + f^{cae} f^{bde} &= 0 & (4)
 \end{aligned}$$

The antisymmetric structure constants f^{abc} determine the nonlinear self-interaction of the gauge field.

The Jacobi identity in equation (4) guarantees consistency of nested commutators.

Gauge-invariant observables depend on traces, Casimir invariants and representation content rather than on an arbitrary matrix basis.

2.2 SU(2): the smallest non-Abelian case

SU(2) has three generators.

In the fundamental representation $T_i = \sigma_i/2$, where σ_i are the Pauli matrices.

Its adjoint representation is three-dimensional and transforms like a spatial vector.

This coincidence makes a diagonal identification between spatial rotations and an internal SU(2) subgroup possible without producing an anisotropic stress at the homogeneous level.

Equations 5-7. SU(2) commutator, quadratic Casimir and representation dimension.

$$[T_i, T_j] = i \epsilon_{ijk} T_k \quad (5)$$

$$C_2(j) = j(j+1) \quad (6)$$

$$\dim(j) = 2j+1 \quad (7)$$

2.3 SU(3), rank two and inequivalent SU(2) embeddings

SU(3) has eight generators, conventionally $T^a = \lambda^a/2$ with the Gell-Mann matrices λ^a .

It has rank two, so two commuting Cartan generators are required.

For cosmology the key issue is not simply that SU(3) has five more generators than SU(2), but that different SU(2) subalgebras can be embedded inequivalently inside SU(3).

The adjoint representation then decomposes differently under the embedded rotational triplet.

Table 2. SU(3) adjoint content recovered by the independent representation audit.

Embedding	Index l_emb	Adjoint decomposition	Cosmological implication
Regular	1	$8 \rightarrow 3 \oplus 2 \oplus 2 \oplus 1$	Triplet plus two doublets and a singlet
Principal	4	$8 \rightarrow 3 \oplus 5$	Triplet plus a spin-2 quintet

The August audit recovered these decompositions numerically, with maximum algebra residual 8.882×10^{-16} and representation gates passing for both embeddings.

Doublet, singlet and quintet modes are therefore the most direct linear candidates for genuinely SU(3)-specific physics.

The triplet alone is generically universal after coupling rescaling.

2.4 Isotropy from a diagonal symmetry

A vector field normally selects a spatial direction.

A non-Abelian gauge multiplet can avoid this because a spatial rotation can be compensated by an internal gauge rotation.

With J_i an embedded SU(2) triplet, the homogeneous ansatz is $A_i = a Q J_i$ and $A_0 = 0$.

The background preserves the diagonal subgroup $SO(3)_{\text{diag}} \subset SO(3)_{\text{space}} \times SU(2)_{\text{gauge}}$.

This is the symmetry reason an isotropic gauge condensate is possible.

3. Gauge Fields in Quantum Cosmology

3.1 Frozen action and field strength

The recent SU(3) programme used a minimal Einstein-scalar-Yang-Mills action as the reference theory.

The pseudoscalar ϕ couples to the topological density $F \cdot \tilde{F}$.

A field-dependent kinetic function $I(\phi)$ was retained in the identifiability audit but set by the selected action in later numerical branches.

Equations 8-10. Reference action, non-Abelian field strength and Hodge dual.

$$S = \int d^4x \sqrt{-g} [M_{\text{Pl}}^2 R/2 - (\partial\phi)^2/2 - V(\phi) - I^2(\phi) F^a_{\mu\nu} F_a^{\mu\nu}/4 - (\lambda/4f)\phi F^a_{\mu\nu} \tilde{F}_a^{\mu\nu}] + S_m \quad (8)$$

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu \quad (9)$$

$$\tilde{F}^a_{\mu\nu} = \epsilon_{\mu\nu\rho\sigma} F^a_{\rho\sigma} / (2\sqrt{-g}) \quad (10)$$

Here M_{Pl} is the reduced Planck mass, R the Ricci scalar, g the gauge coupling, λ/f the pseudoscalar Chern-Simons coupling, $V(\phi)$ the scalar potential and S_m any matter sector.

The parity-odd $F\tilde{F}$ interaction permits one gauge-field helicity to be transiently amplified near horizon crossing, which can source chiral gravitational waves and parity-odd CMB correlations.

3.2 Homogeneous invariants and dimensionless controls

Equations 11-15. Isotropic gauge ansatz and embedding-index invariants.

$$A_i = aQJ_i \quad (11)$$

$$E = Q + HQ \quad (12)$$

$$\mathcal{E} = I_{\text{emb}} E^2 \quad (13)$$

$$\mathcal{B} = I_{\text{emb}} g^2 Q^4 \quad (14)$$

$$\mathcal{P} = I_{\text{emb}} E g Q^2 \quad (15)$$

Equations 16-21. Dimensionless parameters used in the background and perturbation scans.

$$m_Q = gQ/H \quad (16)$$

$$\xi = \lambda\phi/(2fH) \quad (17)$$

$$\epsilon_E = E^2/(H^2 M_{\text{Pl}}^2) \quad (18)$$

$$\epsilon_B = g^2 Q^4/(H^2 M_{\text{Pl}}^2) \quad (19)$$

$$\Omega_A = \rho_A/(3M_{\text{Pl}}^2 H^2) \quad (20)$$

$$x = k/(aH) \quad (21)$$

m_Q measures the gauge-background mass scale relative to:

Hubble expansion;

ξ measures the axion driving;

ϵ_E and ϵ_B are electric and magnetic slow-roll energy fractions;

Ω_A is the total gauge fraction;

and x tracks a Fourier mode from sub-horizon $x \gg 1$ through horizon crossing $x \approx 1$ to super-horizon $x \ll 1$.

3.3 Exact SU(3)-to-SU(2) background mapping

Equations 22-24. Constant-embedding mapping that preserves the homogeneous invariants.

$$Q_2 = \sqrt{I_{\text{emb}}} Q_3 \quad (22)$$

$$E_2 = \sqrt{I_{\text{emb}}} E_3 \quad (23)$$

$$g_2 = g_3/\sqrt{I_{\text{emb}}} \quad (24)$$

Across 256 fixed-seed states per embedding, the largest relative mismatch after equations (22)-(24) was 6.224×10^{-16} .

The logarithmic observable Jacobian had rank two, and the embedding-index direction projected into the existing SU(2) tangent space with residual 4.510×10^{-16} .

Thus the homogeneous-identifiability gate fails in the scientifically informative sense: the minimal background cannot distinguish SU(3) from embedded SU(2).

This agrees with the published SU(N) universality theorem for constant embeddings at tree level.

4. Development of the Research Programme

Table 3. Research history and present evidence status.

Period	Branch	Main question	Current readout
14-16 July	SU(2)/Quaia	Does a $z \approx 1-1.5$ quasar angular mode survive look-elsewhere and survey templates?	Targeted candidate; no global promotion
17 July	SU(2)-BBN/CMB	Can extra SU(2)-like radiation or the published thermal bridge solve lithium?	No joint D+He+Li pass; identities kept separate
17 August	SU(3) algebra/background	Are regular/principal embeddings distinguishable and dynamically occupied?	Background-equivalent; regular basin replicated
17-18 August	Backreaction/phase maps	Where are topology-loss and extinction boundaries?	Two distinct empirical boundary curves resolved
18 August	Finite-k S/V/T	Do qualified backgrounds pass perturbative stability and freeze-out?	Tensor technical pass; scalar/vector gates fail
18-19 August	Exit/reheating	Can perturbations be propagated through a physical exit and microscopic portal?	Numerically controlled but action/operator sensitive
19 August	Consistency audit	Do code or null-construction defects reverse the conclusions?	30/30 SU(3) controls pass; SU(2) nulls corrected; no promotion

5. SU(2) Observational Programme

5.1 Quaia redshift-dipole search

Quaia combines Gaia DR3 low-resolution spectrophotometric quasar candidates with unWISE infrared photometry and SDSS-trained redshift correction.

The public $G < 20.0$ catalogue contains 755,850 objects.

The corrected coherent-null analysis used 668,558 objects after the frozen mask and quality preparation, scanning 168 overlapping redshift-window and Galactic-latitude combinations.

The targeted family was $z = 1.0-1.5$ across $|b|$ cuts from 10° to 35° .

Equations 25-28. Angular redshift-dipole fit, Wald statistic and empirical global probability.

$$z_i = c_0 + d \cdot n_i + \varepsilon_i \quad (25)$$

$$A_{\text{dip}} = ||d|| \quad (26)$$

$$S^2 = d^T \text{Cov}(d)^{-1} d \quad (27)$$

$$p_{\text{emp}} = [1 + \#\{T_m \geq T_{\text{obs}}\}]/(M+1) \quad (28)$$

The initial all-window code generated an independent random sky for each overlapping window and therefore broke catalogue-level correlations.

The August audit corrected this by drawing one coherent null catalogue per mock and passing it through all windows.

For the historical strongest $|b| > 35^\circ$ row, the maximum-SNR global probability changed from 0.432713 to 0.249169; the maximum-priority probability changed from 0.200760 to 0.106312; and the targeted-family probability was 0.016611.

The calibration changed materially, but the global result remained ordinary.

5.2 Matched catalogue quality and scan-law/colour structure

A dipole can be induced or distorted if catalogue quality varies across the fitted hemispheres.

The matched-quality analysis estimated a propensity for hemisphere membership from redshift error, G magnitude, Gaia BP-RP, $W1$, $W2$, $W1-W2$ and proper-motion uncertainties, then used overlap weights.

Under the all-quality model the dipole amplitude fell to 0.309709 of baseline, the axis rotated by 121.39°, and the maximum SNR fell to 1.87638 despite excellent weighted balance.

This keeps catalogue quality as the leading explanation.

Equations 29-32. Joint dipole/nuisance likelihood and overlap weights.

$$z_i = \alpha_w + d_w^T n_i + \beta_w^T C_i + \varepsilon_i \quad (29)$$

$$\varepsilon_i = \text{Normal}(0, \sigma_w^2/w_i) \quad (30)$$

$$w_i(+) = 1 - \hat{p}_i \quad (31)$$

$$w_i(-) = \hat{p}_i \quad (32)$$

The historical scan-law/colour residual search returned $p=0.000999$, but that number was not a calibrated catalogue-level global probability because residuals were permuted independently in overlapping windows.

A later 50,000-mock spatial-block audit found family probabilities 0.013520, 0.092298 and 0.232595 for 8°, 12° and 16° blocks.

Dependence on block scale prevents physical promotion and is exactly what a survey-template explanation predicts.

5.3 Coherent joint-nuisance recalibration

The latest production run refits the complete scan-law/colour nuisance model, mock-specific dipole hemisphere, quality propensity, overlap weights and final weighted likelihood inside every coherent catalogue mock.

Its report status is COHERENT_JOINT_NUISANCE_PROMOTION_NOT_PASSED with 300 mocks.

For $z=1.0-1.5$ and $|b|>10$ deg, the matched statistic is $\text{SNR}=4.42293$ with amplitude 0.00279036 and axis $(l,b)=(229.434 \text{ deg}, 4.15561 \text{ deg})$.

The calibrated probabilities are global SNR $p=0.0398671$, targeted-family SNR $p=0.00332226$, global priority $p=0.0166113$ and targeted-family priority $p=0.00332226$.

The targeted family reaches the finite-mock floor 1/301, but both all-window probabilities exceed 0.01.

The preregistered promotion rule therefore fails even before enforcing direction and amplitude stability across latitude cuts.

Interpretation of the Quaia mode

The $z \approx 1-1.5$ feature is useful because it concentrates the systematics question into a fixed redshift range and direction.

It is not yet useful as evidence for SU(2).

The decisive next test is independent cross-catalogue replication with the same mask, windows, spatial-block null and survey-template likelihood preregistered before looking at the result.

5.4 HHT and primordial-abundance controls

A Hilbert-Huang transform (HHT) surrogate analysis was repaired so that seeds no longer depend on Python's process-randomised hash function.

Two independent 500-shuffle runs became byte-identical.

Stable top-energy probabilities were 0.738523 for amplitude projected on the CMB direction, 0.491018 for the fractional projection and 0.714571 for dipole amplitude: all ordinary.

The effective SU(2)-compatible Big-Bang nucleosynthesis (BBN) gate scanned 30,503 LINX points.

None passed deuterium, helium-4 and lithium-7 simultaneously.

A separate high-temperature SU(2)-CMB asymptotic bridge produced $N_{\text{eff}}=20.373$ and failed abundance gates.

These results do not exclude every SU(2) thermal theory; they exclude the implemented expansion-only/eight-mode mappings and demonstrate why a gauge-matter action, entropy transfer and nuclear-response model are required.

6. SU(3) Background Dynamics and Phase Structure

6.1 Non-perturbative basin map

The first broad initial-condition ensemble integrated 528 cases over $\xi=\{2.2,2.6,3.0,4.0\}$.

Excluding exact controls, 117 trajectories ended in the regular embedding, so the regular-basin replication gate passed.

Endpoint counts were 206 non-embedding/stress-gate, 177 principal, 121 regular and 24 vanishing-field.

A dense low- ξ refinement then integrated 756 cases and resolved 97 adjacent branch-transition brackets to $\Delta t=0.025$.

These are classical topology results on a frozen background, not primordial spectra.

6.2 Backreaction and the two-boundary phase map

When selected cells were coupled to axion, Friedmann and shear backreaction, only 12/17 weak-regime endpoints survived and two of three preregistered $P \rightarrow R \rightarrow V \rightarrow R$ ordering gates failed.

Stronger coupling or elevated gauge energy drove all 17 selected active condensates to the vanishing-field branch.

This prevented a simple claim that the fixed-background topology is robust.

Equations 33-34. Empirical topology-loss and extinction boundaries for the frozen coupled action.

$$\log_{10}\Omega_{\text{loss}} = 0.037760 - 3.507679 \log_{10}\kappa \quad (33)$$

$$\log_{10}\Omega_{\text{ext}} = -0.587635 - 2.231578 \log_{10}\kappa \quad (34)$$

Adaptive refinement added 192 trajectories and bracketed both edges for $\kappa=10, 15, 25$ and 50.

Equations (33)-(34) show that topology loss and complete extinction are distinct curves enclosing a finite mixed phase.

Their slopes are not universal critical exponents; they are high-quality empirical summaries of this action and case design.

Only points comfortably below Ω_{loss} are eligible for perturbation calculations.

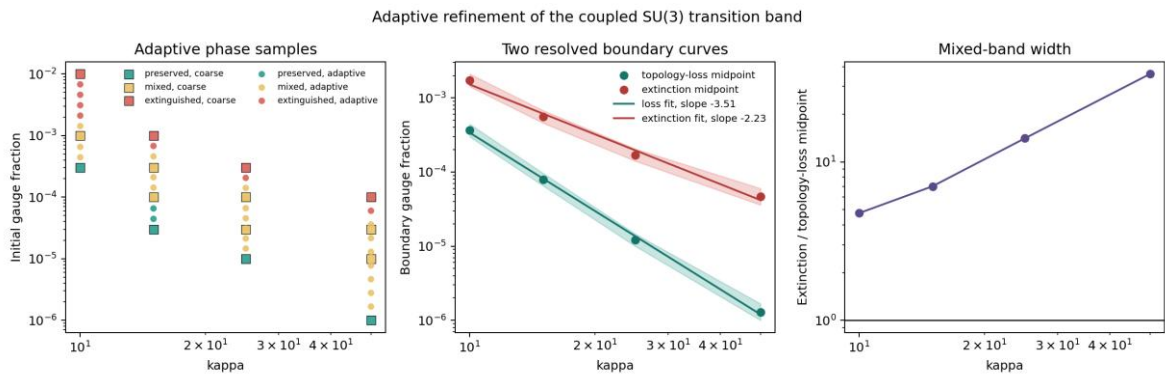


Figure 1. Adaptive refinement of the coupled SU(3) transition band. The centre panel separates topology-loss and extinction curves; the widening mixed band demonstrates that loss of branch ordering precedes complete condensate extinction.

Source: su3_kappa_omega_adaptive_refinement_2026-08-17.png

6.3 Long-duration anchors and identifiability

Interior anchors K10, K15, K25 and K50 were placed about a factor of ten below the lower topology-loss bracket.

K10 and K15 remained sufficiently qualified for finite-k work; K25/K50 initially failed parts of the long-background gate.

Scalar and vector triplet transfers agreed between regular and principal embeddings after $g \rightarrow g\lambda$ normalisation, as predicted by linear universality.

The tensor self-transfer decayed beneath the numerical floor, making a relative leakage statistic ill-conditioned rather than demonstrating physical growth.

7. Finite-Wavenumber Perturbations

7.1 Helicity-resolved tensor sector

For helicity $h=\pm 1$, the canonical gauge-tensor frequency and coupled metric/gauge equations were evolved from $x=100$ to $x=0.01$.

The kinetic and high-k gradient matrices were the identity.

The $h=+1$ gauge tensor experienced a finite tachyonic interval near horizon crossing; $h=-1$ did not.

Equations 35-37. Canonical finite-k tensor system.

$$\omega^2_{T,h} = 1 + 2m_Q \xi / x^2 - 2h(m_Q + \xi) / x \quad (35)$$

$$\Psi_{h,xx} + (1 - 2/x^2)\Psi_h = 2\sqrt{\epsilon_E} T_{h,x}/x + 2\sqrt{\epsilon_B}(m_Q - hx)T_h/x^2 \quad (36)$$

$$T_{h,xx} + \omega^2_{T,h} T_h = -2\sqrt{\epsilon_E} \Psi_{h,x}/x + 2[\sqrt{\epsilon_B}(m_Q - hx) + \sqrt{\epsilon_E}] \Psi_h/x^2 \quad (37)$$

For the K10/K15 anchors the largest control-normalised late metric excess was 53.695 parts per million (ppm) and the largest absolute chirality was 22.067 ppm.

This is a technically successful chiral transfer calculation, but not a detectable prediction: both anchors had $m_Q < \sqrt{2}$ and the scalar/vector sectors were not yet qualified.

The later scalar work refined this warning by showing that the familiar threshold is neither necessary nor sufficient for all super-horizon behaviour in the local parameter regime.

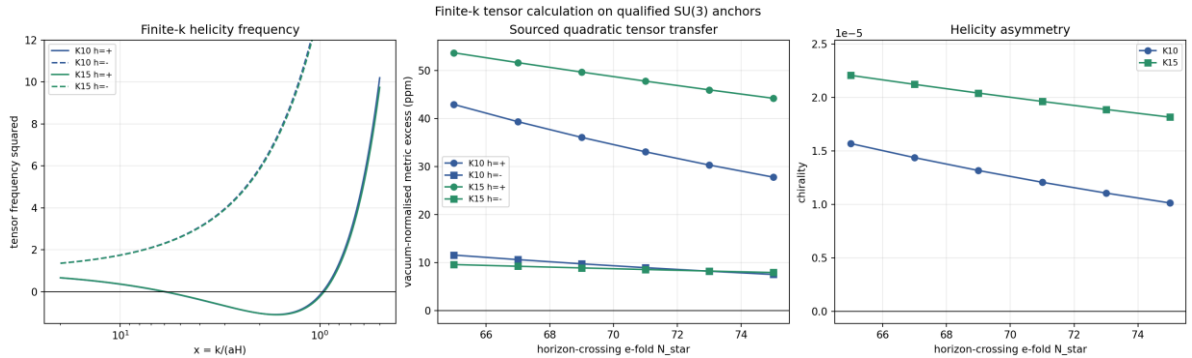


Figure 2. Finite-k tensor readout for K10/K15 anchors. Only the positive helicity develops the transient tachyonic interval; the resulting metric chirality remains at the tens-of-ppm level for these energetically subdominant backgrounds.

Source: su3_finite_k_tensor_readout_2026-08-18.png

7.2 Constraint-reduced scalar action

The complete scalar calculation retained lapse, scalar shift and temporal gauge constraints and eliminated them by a numerical Schur complement.

For dynamical coordinates q and nondynamical constraints c :

Equations 38-41. Constraint elimination by Schur complement.

$$c = -T^{-1}(Rq' + Sq) \quad (38)$$

$$A_{\text{eff}} = A - R^T T^{-1} R \quad (39)$$

$$B_{\text{eff}} = B - R^T T^{-1} S \quad (40)$$

$$C_{\text{eff}} = C - S^T T^{-1} S \quad (41)$$

All K10/K15 modes passed kinetic, gradient, constraint and convergence tests, yet all failed the curvature-freeze gate between $x=0.1$ and $x=0.01$.

Freeze drifts ranged from about 216 to 629 for $\xi=2.08$.

Moving to $\xi=2.15$ raised m_Q above $\sqrt{2}$ but worsened the drift to roughly 3.1×10^3 - 7.5×10^3 .

This unexpected result is important: $m_Q > \sqrt{2}$ removes the conventional rapid sub-horizon instability but does not guarantee a conserved adiabatic curvature mode.

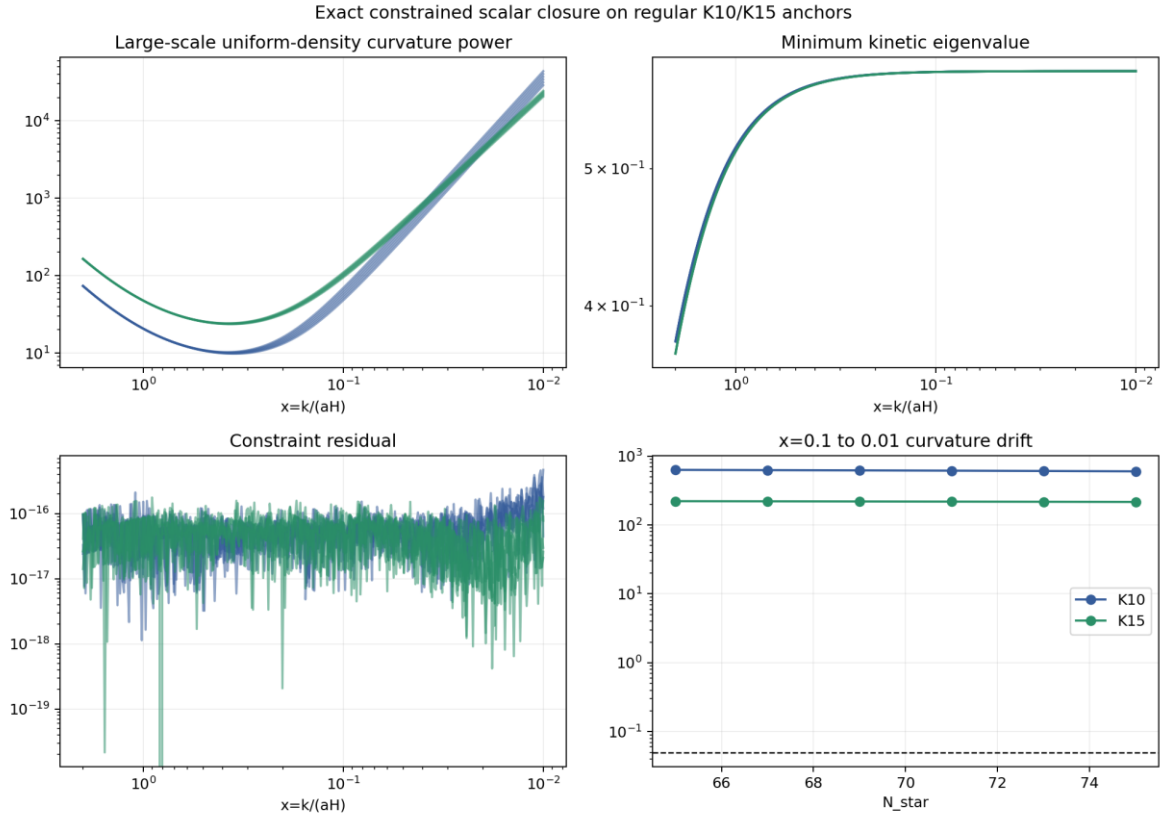


Figure 3. Full constrained scalar closure. Positive kinetic and gradient eigenvalues coexist with a failed curvature-freeze gate, demonstrating that numerical health and physical adiabaticity are distinct requirements.

Source: su3_full_scalar_closure_readout_2026-08-18.png

7.3 Low- ξ K25 basin and scalar-vector-tensor split

A targeted low- ξ branch at $\xi_0=2.002158352$, $\kappa=25$ and $\Omega_{A,0}=10^{-5}$ passed a direct 51/51 core initial-condition basin ensemble.

Nevertheless, every scalar epoch failed curvature freeze with drift about 1.29-1.33, and the positive-helicity vector mode reached canonical power about 13.6-13.8.

The tensor sector remained technically controlled, with positive-helicity metric power only about 0.11% above vacuum.

The result is therefore LOWXI_BASIN_PASS_FINITEK_NOT_PASSED.

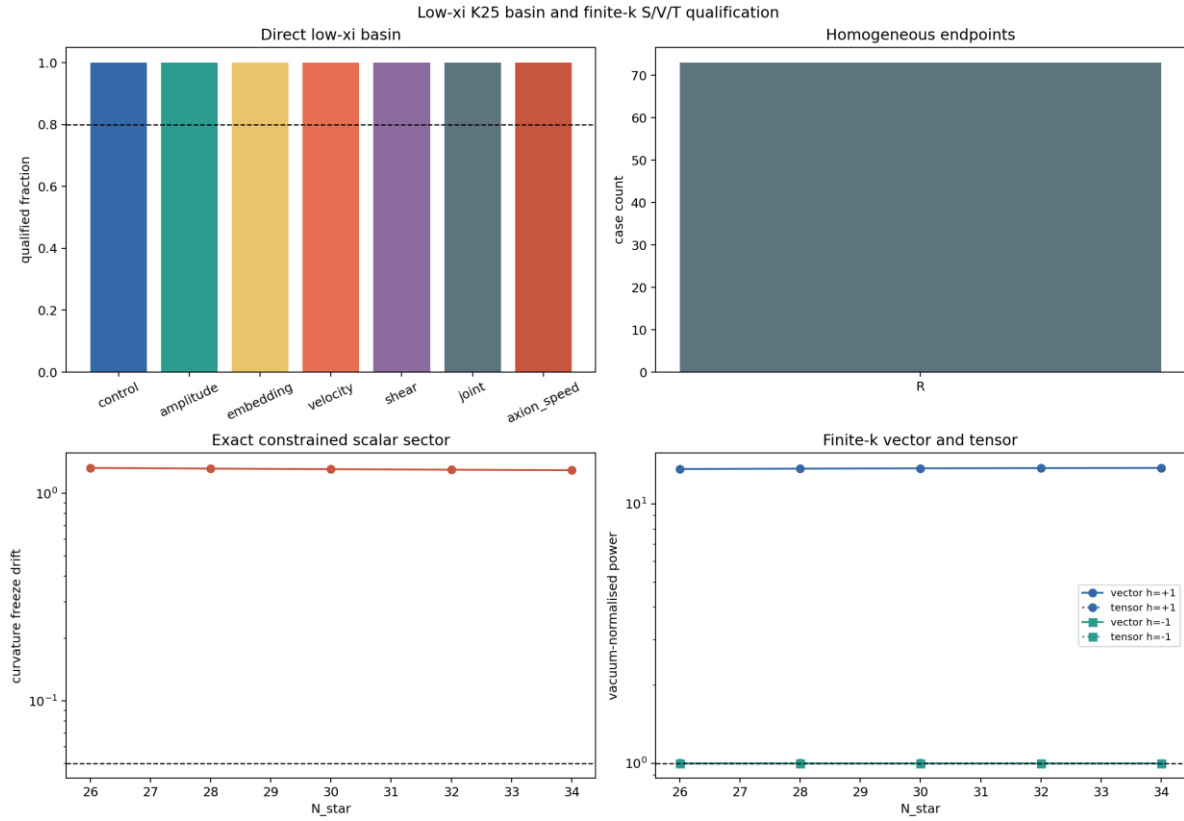


Figure 4. Low- ξ K25 qualification. The homogeneous basin passes, while the scalar freeze and positive-helicity vector gates fail; the tensor response remains small.

Source: *su3_lowxi_basin_finitek_svt_readout_2026-08-18.png*

7.4 Eigenmode-resolved super-horizon diagnosis

Extending the exact scalar transfer from $x=0.01$ to $x=10^{-4}$ showed continued but slowing evolution.

At $N^*=30$ the two dominant coherent allocations were $+8.825798$ and -7.825735 times the final curvature power.

Their physical exponents were approximately $+1.45 \times 10^{-5}$ and -0.01146 after removing the $p=aq$ scale-factor exponent.

The observed growth is therefore relaxation of destructive interference between nearly constant axion-dominated modes, not a ghost or explosive local eigenmode.

This refined interpretation matters.

A failed freeze gate can mean:

- (i) a true tachyonic instability,
- (ii) a slowly decaying isocurvature source,
- (iii) a poorly chosen hypersurface, or
- (iv) destructive interference that has not reached its asymptote.

The eigenvector audit supports mechanism (iv), while still withholding observational promotion because the late conserved curvature mode has not been demonstrated.

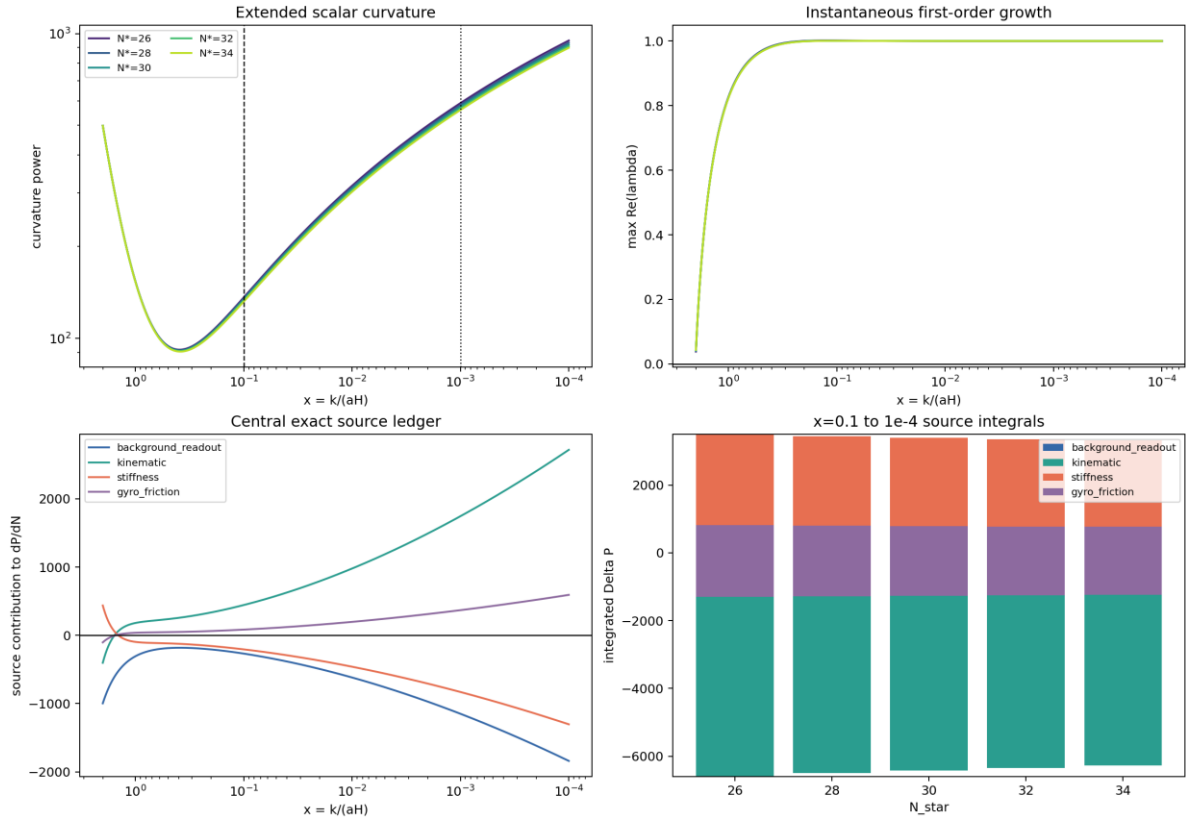


Figure 5. Eigenmode-resolved scalar continuation to $x=10^{-4}$. Continued evolution is traced to cancellation between two axion-dominated modes rather than a negative kinetic eigenvalue.

Source: *su3_lowxi_scalar_eigenmode_audit_readout_2026-08-18.png*

8. Exit, Reheating and Quantum-Channel Closure

8.1 Why reheating is not an optional afterthought

Primordial perturbations become observable only after the inflationary degrees of freedom transfer energy into visible and possibly dark fluids.

An imposed end surface can hide strong dependence on end time or entropy transfer.

In the low- ξ branch, a conditional instantaneous single-bath adapter was numerically precise, but the final radiation-curvature power varied by up to 95.3% when the imposed end moved from $N=55$ to $N=65$.

A later smooth exit generated $\varepsilon=1$ dynamically, but the direct propagator failed the preregistered 0.1% precision gate because of enormous dynamic range.

8.2 Continuous orthonormalisation and decay operators

Segmented complex-QR propagation kept the growing fundamental matrix finite and orthonormal.

The parent 6,401-point matrix grid missed its precision threshold by 1.696×10^{-3} ; a post-hoc 25,601-point refinement passed at 7.324×10^{-4} but did not overwrite the preregistered parent decision.

This illustrates the project's rule that post-hoc numerical success informs the next run but does not retroactively change a frozen gate.

Equations 42-45. Background energy-transfer model used to test explicit visible and dark decay branches.

$$\rho'_{\varphi} = -3\rho_{\varphi} - (\Gamma_{\varphi v}/H)\rho_{\varphi} \quad (42)$$

$$\rho'_{\text{A}} = -4\rho_{\text{A}} - [(\Gamma_{\text{Av}} + \Gamma_{\text{Ad}})/H]\rho_{\text{A}} \quad (43)$$

$$\rho'_{\text{v}} = -4\rho_{\text{v}} + (\Gamma_{\varphi v}/H)\rho_{\varphi} + (\Gamma_{\text{Av}}/H)\rho_{\text{A}} \quad (44)$$

$$\rho'_{\text{d}} = -4\rho_{\text{d}} + (\Gamma_{\text{Ad}}/H)\rho_{\text{A}} \quad (45)$$

8.3 Microscopic portal audit

Equations 46-49. Candidate dimension-five portals and first-order perturbed energy conservation.

$$\mathcal{L}_{\text{portal}} = -(g_{\phi v}/4)\phi F_v \tilde{F}_v - (g_{Av}/4)\chi_{AF_v} F_v - (g_{Ad}/4)\chi_{AF_d} F_d \quad (46)$$

$$\nabla_\mu T_i^{\mu\nu} = Q_i^{\nu} \quad (47)$$

$$\Sigma_i Q_i^{\nu} = 0 \quad (48)$$

$$\delta Q_{\phi v} = \Gamma_{\phi v} \delta \rho_{\phi} + \rho_{\phi} \delta \Gamma_{\phi v} \quad (49)$$

Six of seven portal choices passed the effective-field-theory control; the fast-scalar case failed because $g_{\phi v} m_{\phi} \approx 2.84$ exceeded the perturbative bound.

All seven super-horizon tangent and conservation checks passed.

Yet the final entropy fraction ranged from 0 in the no-dark null to 0.594806 in the gauge-dark case.

Numerical closure is therefore not predictive closure: the parent action must select the portal coefficients rather than fitting them after inspecting the desired outcome.

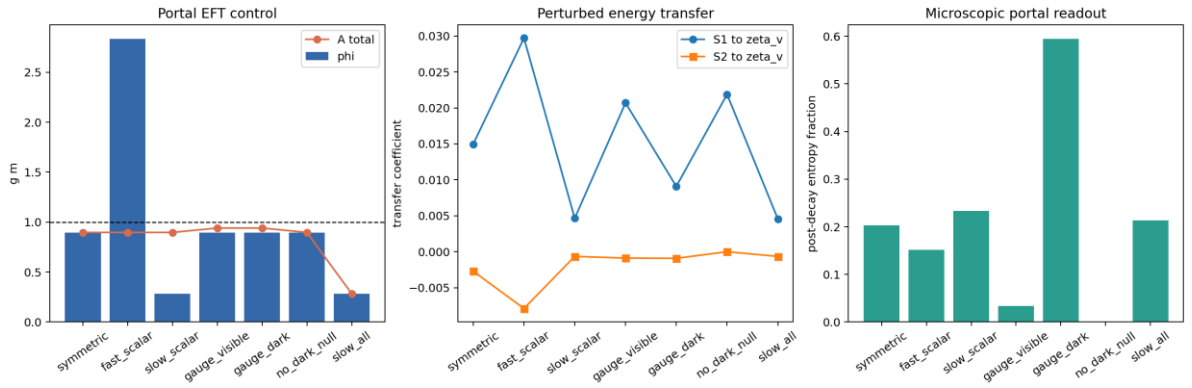


Figure 6. Microscopic portal and perturbed energy-transfer audit. Conservation and tangent checks pass, but the post-decay entropy fraction remains strongly operator-dependent.

Source: su3_microscopic_reheating_readout_2026-08-19.png

9. Cosmology Datasets Used

Table 4. Observational and computational datasets already used.

Dataset or product	Role in the programme	Scale/status	Principal caveat
Quaia G<20.0	All-sky quasar redshift-dipole and systematics tests	755,850 public objects; 668,558 in coherent prepared sample	Photometric/spectrophotometric redshift and angular selection systematics
Gaia DR3	Sky positions, BP/RP colour, scan counts/geometry, astrometry and quasar candidates	Space-based all-sky catalogue	Scanning law, calibration and colour-dependent completeness
unWISE/WISE	W1/W2 colours and quasar purity control	All-sky infrared photometry	Depth, dust and stellar contamination
SDSS redshift training/cross-match	Reference spectroscopy for Quaia redshift correction and independent-catalogue design	Ground-based spectroscopic sample	Footprint and selection differ from Quaia
Planck CMB products/dipole	Directional control and target for future TB/EB, BB and lensing likelihoods	Full-sky satellite maps; PR4/NPIPE available	Foregrounds, masks, calibration-angle degeneracy
LINX + PRIMAT rates	BBN expansion, entropy and abundance propagation	30,503 completed effective-SU(2) points	Not a first-principles gauge-matter nuclear model
Synthetic SU(3) ensembles	Basin, backreaction, phase, finite-k and reheating qualification	Thousands of preregistered trajectories/modes	Theory-generated, not observational data
Adjacent project data: SPARC, Pantheon+/DES/Union, DESI BAO	Galaxy rotation and distance-sector controls in the wider cosmology programme	Used outside the present SU(3) likelihood	Must not be cited as SU(3) evidence without an observable adapter

10. Extended and Satellite-Derived Data Programme

The most useful future datasets are those that address a specific open gate.

The table below distinguishes data already public from releases that were upcoming on 19 August 2026.

Availability dates should be rechecked immediately before a proposal or analysis is frozen.

Table 5. Available and prospective data matched to the open theory gates.

Facility/release	Availability at 19 Aug 2026	Relevant observable	Priority use
Planck PR4/NPIPE	Public	Full-sky T/E/B maps, low- ℓ anomalies, lensing	TB/EB parity, chirality and SU(2)/SU(3) transfer likelihood
ACT DR6/DR6.02	Public	90/150/220 GHz T/P maps, spectra, lensing and simulations	Independent high- ℓ and lensing cross-check against Planck
Quaia + public full-sky mocks	Public catalogue; newer tailored mocks available	Quasar angular clustering and redshift systematics	Replace empirical shuffles with end-to-end selection-function mocks
Gaia DR4	ESA lists as upcoming	66-month epoch/scan-law products, ~2.7 billion processed sources	Rebuild Quaia-like systematics model from lower-level observations
Euclid Q1/Q2	Q1 public; Q2 listed June 2026	Deep-field imaging/spectroscopy, morphology, weak-lensing preparation	Cross-catalogue angular and morphology/environment controls; Q1 is not a core cosmology release
Euclid DR1	Foundation release listed for Nov 2026	Wide-area shapes, clustering and spectroscopy	Held-out sky, lensing and large-scale-structure transfer tests
Rubin EDP2	Released 27 July 2026	LSSTCam imaging and time-domain products	Independent quasar selection, depth/scan templates and transient controls
Roman	Launch scheduled 30 Aug 2026	Wide/deep infrared imaging, spectroscopy, SNe and weak lensing	Future high-redshift transfer, lensing and source-evolution tests
LiteBIRD	Future; JAXA targets JFY2036	Full-sky multi-frequency CMB polarisation	Definitive large-scale BB and parity/chirality constraints

Facility/release	Availability at 19 Aug 2026	Relevant observable	Priority use
LISA	Future; ESA launch planned 2035	0.1 mHz-0.1 Hz gravitational-wave background	Blue/chiral primordial spectra and phase-transition tests

11. Observable Adapters and Falsifiable Predictions

11.1 CMB scalar, vector and tensor spectra

A Boltzmann code requires primordial covariance spectra $P_{IJ}(k)$ on a specified post-reheating hypersurface, not only background parameters.

The minimum adapter must output adiabatic curvature, each surviving isocurvature mode and both tensor helicities.

Schematically:

Equations 50-53. Primordial covariance, tensor chirality and CMB angular spectra.

$$P_{IJ}(k) = (k^3/2\pi^2)\langle X_I(k)X_J^*(k) \rangle \quad (50)$$

$$P_T = P_+ + P_- \quad (51)$$

$$\chi_T = (P_+ - P_-)/(P_+ + P_-) \quad (52)$$

$$C_{\ell}^{\{XY\}} = 4\pi \int d\ln k \Sigma_{IJ} \Delta_{\ell}^{\{X,I\}} P_{IJ} \Delta_{\ell}^{\{Y,J\}} \quad (53)$$

A parity-symmetric primordial model has vanishing TB and EB in the absence of rotation or instrumental leakage.

A chiral gauge source can generate non-zero TB/EB, but cosmic birefringence, polarisation-angle calibration, Galactic dust and beam leakage are direct competitors.

The likelihood must therefore fit the gauge signal and these nuisance rotations jointly, with frequency and sky-split validation.

11.2 Large-scale structure and lensing

If the non-Abelian sector survives beyond inflation or changes the primordial scalar covariance, it can affect matter growth, CMB lensing, galaxy clustering and cross-correlations.

A robust test should propagate perturbations through a modified Einstein-Boltzmann solver and compare Planck PR4, ACT DR6 lensing, Quaia, Euclid and Rubin with a common survey nuisance model.

A late-time SU2/SU2R fluid must be derived from the same action before it is combined with the inflation branch.

11.3 Gravitational-wave frequency ladder

CMB B modes probe the largest scales; pulsar timing arrays probe nanohertz frequencies; LISA probes millihertz; terrestrial interferometers probe tens to thousands of hertz.

A time-dependent gauge activation can produce a blue or localised tensor feature.

The present tens-of-ppm K10/K15 response is not a forecast, but the machinery can be extended to a scale-dependent $\xi(N)$ and mapped to today through the reheating equation of state.

The chirality fraction in equation (52), spectral tilt and tensor bispectrum are more diagnostic than total amplitude alone.

12. Interesting and Unexpected Results

Table 6. Novel findings, disciplined interpretation and routes by which genuinely new physics could emerge.

Result	Why it is interesting	Conservative inference	Possible new-physics route
Exact homogeneous SU(3)/SU(2) equivalence	More generators do not automatically create a new background observable	Expansion-history fitting cannot identify the gauge group	Embedding transitions, extra representations, loops or confinement
Finite regular basin plus two phase boundaries	Topology loss precedes complete field extinction	Initial conditions and gauge energy constrain viable histories	Transition-generated localised GW/CMB feature
$m_Q > \sqrt{2}$ branch still fails freeze	Standard sub-horizon stability criterion is insufficient	Super-horizon adiabaticity needs its own boundary	Modified friction, potential, kinetic structure or entropy conversion
Low- ξ basin passes but vector/scalar fail	Background robustness does not predict perturbative viability	Do not promote homogeneous attractors	Screening or alternative operators that alter only unstable sectors
Scalar growth is interference relaxation	Failure is not a ghost or runaway eigenvalue	Longer propagation and physical exit are mandatory	Observable correlated isocurvature if reheating freezes the mixture
Portal transfer passes but is operator-sensitive	Numerical consistency does not select couplings	No unique abundance/entropy prediction yet	UV completion or symmetry may fix portal ratios
Quaia family-local $p=1/301$ but global gates fail	Nuisance relearning preserves a localised candidate while look-elsewhere control blocks promotion	No SU(2) detection	Only independent-catalogue, held-out-sky replication could reopen promotion

13. Priority Research Paths

Repeat the frozen $z=1.0-1.5$ search in an independent quasar catalogue, using the completed coherent joint-nuisance calibration as the baseline.

The same windows, masks, latitude cuts, spatial-block scales and global statistic must be locked before inspection; a 5,000-mock extension is warranted only if the independent catalogue reproduces the mode.

Build the genuinely SU(3)-specific quadratic action for the regular doublet+doublet+singlet and principal quintet sectors.

Verify kinetic and gradient eigenvalues and calculate their coupling, if any, to curvature, tensors or reheating at nonlinear order.

Replace the phenomenological exit and portal menu with one symmetry-selected potential $V(\phi)$, kinetic/friction sector and visible/dark interaction Lagrangian.

Recompute the background, finite- k transfer and reheating covariance without retuning gates.

Implement a continuously orthonormalised high-precision propagator with logarithmic amplitudes, independent integrators and a frozen convergence hierarchy.

Require the parent production run, not only a post-hoc refinement, to pass.

Export P_{RR} , P_{RS} , P_{SS} , P_+ , P_- and their phases on the post-reheating uniform-density surface to CLASS or CAMB.

Begin with Planck PR4 and ACT DR6 null/likelihood validation before adding Euclid, Rubin or Roman.

Run an injection-calibrated TB/EB/BB likelihood that jointly fits cosmic birefringence, polarisation-angle calibration, dust, synchrotron and beam leakage.

Use held-out frequencies and sky regions as promotion gates.

Test transition histories between regular and principal embeddings.

Search for a localised, scale-dependent chiral tensor feature rather than a constant-amplitude template; forecast across CMB, PTA, LISA and terrestrial bands.

Keep the high-temperature SU(2)-CMB, late-time SU2/SU2R and axion-gauge branches separate until an explicit action and entropy history maps one into another.

Shared notation is not sufficient.

Table 7. Criticality matrix for the next research phase.

Priority	Gate	Minimum success criterion	Stop/redirect criterion
P1	Independent Quaia replication	Global and targeted $p \leq 0.01$ with stable amplitude/direction after full nuisance relearning	Quality/template model wins or block-scale instability persists
P1	SU(3)-specific modes	Positive kinetic and c_s^2 eigenvalues plus non-zero observable Jacobian after SU(2) projection	Modes decouple or remain exactly universal
P1	Action-selected exit/portals	Unique couplings and $<0.1\%$ transfer convergence	Outcome changes materially across equally allowed operators
P2	CMB parity likelihood	Injection-calibrated TB/EB/BB recovery and held-out-frequency prediction	Calibration/birefringence/foreground model explains signal
P2	Lensing/growth	Common Planck/ACT/Quaia/Euclid likelihood improves predictive score	No consistent cross-probe transfer
P3	Multi-band GW forecast	Scale-dependent spectrum survives CMB and BBN bounds	Required amplitude violates backreaction or scalar constraints

14. Limitations and Falsification Boundaries

- Minimal-action limitation.** Most SU(3) calculations use a canonical Einstein-pseudoscalar-Yang-Mills action or a documented Horndeski-friction extension.
A different kinetic or matter-coupling sector can alter the boundaries.
- Embedding limitation.** The strongest finite-k results concern the universal embedded SU(2) triplet.
They do not yet solve the extra regular doublets/singlet or principal quintet.
- Linear limitation.** Tree-level equivalence does not cover loops, nonlinear mode coupling, fragmentation, confinement or thermalisation.
- Reheating limitation.** Portal widths are candidate effective operators, not a UV-complete derivation.
Operator sensitivity is part of the result.
- Survey limitation.** Quaia angular structure is exposed to Gaia scanning geometry, colour selection, infrared depth, dust, stellar contamination and redshift training.
Coherent end-to-end mocks are essential.
- Statistical limitation.** Three hundred coherent mocks resolve probabilities only down to $1/301$.
A 5,000-mock confirmation is appropriate for publication-level tails but is not needed to preserve a clear non-detection.
- Identity limitation.** SU(2)-CMB thermodynamics, late-time SU2/SU2R phenomenology and inflationary axion-gauge dynamics remain separate until one action derives their common observables.

What would falsify the current model branch?

A robust falsification would be a failure of the complete constrained action under independent implementation;
a ghost or unacceptable gradient instability in every basin-qualified branch;
a scalar/vector instability that cannot be screened without violating EFT or observations;
an exit/reheating prediction that cannot converge;
or observational spectra excluded by Planck/ACT/BBN after all parameters are fixed independently.

Conversely, a local Quaia p-value or a background basin pass is not sufficient confirmation.

15. Conclusions

The recent programme has moved from loose group labels to a disciplined chain from Lie algebra to observables.

SU(3) regular and principal embeddings are implemented accurately;

their homogeneous invariants are:

- SU(2)-equivalent after embedding-index rescaling;
- broad and dense ensembles reveal real basin topology;
- backreaction separates topology loss from extinction;
- and finite-k calculations isolate tensor, vector and scalar behaviour.

The most useful scientific result is not a detection but a map of where plausible shortcuts fail.

Three findings deserve emphasis.

First, the gauge group is not identifiable from a homogeneous expansion curve under the minimal constant-embedding action.

Second, a stable background and even $m_Q > \sqrt{2}$ do not guarantee a conserved curvature perturbation.

Third, a numerically closed reheating transfer does not become predictive until the action selects its portal operators.

These are refinements of interpretation that prevent false positives and identify exactly where new physics must enter if the framework is to survive.

The observational SU(2) branch is similarly constrained.

The Quia $z \approx 1-1.5$ feature remains an interesting survey-sensitive target, but corrected global nulls, catalogue-quality matching and block-scale dependence prevent promotion.

The next decisive work is therefore not a larger unfocused scan.

It is an independent catalogue replication, an SU(3)-specific perturbation calculation, an action-selected exit/reheating model and a CMB likelihood built from fully propagated primordial covariance spectra.

Appendix A. Symbols and Variables

Table A1. Symbols used in the text.

Symbol	Meaning
a	Scale factor
A^a_{μ}	Gauge potential with adjoint index a and spacetime index μ
$C_{\ell}^{\{XY\}}$	CMB angular cross-power spectrum between observables X and Y
c_s^2	Eigenvalue of the high- k gradient/kinetic matrix; squared propagation speed
d	Three-component fitted angular dipole vector
E	Homogeneous gauge electric amplitude $\dot{Q}+HQ$
f	Axion decay constant
f^{abc}	Lie-algebra structure constants
$F^a_{\mu\nu}$	Non-Abelian field-strength tensor
$\tilde{F}^a_{\mu\nu}$	Dual field-strength tensor
g	Gauge coupling
$g_{\phi\nu}, g_{Av}, g_{Ad}$	Candidate visible/dark portal couplings
H	Hubble expansion rate
h	Helicity label, +1 or -1
$l(\phi)$	Gauge kinetic function
l_{emb}	Index of the embedded SU(2) subalgebra
J_i	Embedded SU(2) generators aligned with spatial axes
k	Comoving wavenumber
κ	Dimensionless scalar/gauge feedback parameter used in the local action scan
λ	Dimensionless Chern-Simons coupling or, in cited universality work, an embedding normalisation; context distinguishes them
m_Q	Gauge-background mass parameter gQ/H
M_{Pl}	Reduced Planck mass
n_i	Unit sky-direction vector of catalogue object i
N	Number of e-folds
N^*	Mode reference epoch in e-folds
Ω_A	Gauge-field fractional energy density
Ω_{loss}	Empirical topology-loss boundary
Ω_{ext}	Empirical complete-extinction boundary
P_{IJ}	Primordial covariance spectrum for modes I and J
P_R, P_S	Adiabatic-curvature and isocurvature powers
Q	Homogeneous gauge amplitude
$Q_i^{\mu\nu}$	Energy-momentum transfer four-vector into component i
R	Ricci scalar; also regular embedding when used as a branch label
S	Wald SNR or isocurvature mode, according to context
T^a	Lie-algebra generator
V	Scalar potential; also vanishing-field endpoint in branch labels

Symbol	Meaning
w_i	Overlap weight in the matched-quality likelihood
x	Dimensionless scale $k/(aH)$
ξ	Axion driving parameter $\lambda\dot{\phi}/(2fH)$
χ_T	Tensor chirality fraction
ϕ	Pseudoscalar/axion field
ρ_i	Energy density of component i
Γ	Decay width
ϵ_E, ϵ_B	Electric and magnetic gauge slow-roll fractions
ϵ_i	Catalogue residual or slow-roll parameter where explicitly defined
ζ	Curvature perturbation on uniform-density hypersurfaces

Appendix B. Acronyms

Table B1. Acronyms and abbreviations.

Acronym	Meaning
ACT	Atacama Cosmology Telescope
ADQL	Astronomical Data Query Language
BAO	Baryon acoustic oscillation
BB	CMB B-mode auto-spectrum
BBN	Big-Bang nucleosynthesis
BIC	Bayesian information criterion
CAMB	Code for Anisotropies in the Microwave Background
CLASS	Cosmic Linear Anisotropy Solving System
CMB	Cosmic microwave background
DESI	Dark Energy Spectroscopic Instrument
DOP853	Eighth-order explicit Runge-Kutta integrator with embedded error control
DR	Data release
EB/TB	Parity-odd CMB E-B and temperature-B cross-spectra
EFT	Effective field theory
ELPD	Expected log predictive density
ESA	European Space Agency
HHT	Hilbert-Huang transform
JAXA	Japan Aerospace Exploration Agency
LINX	Differentiable Big-Bang nucleosynthesis framework
LISA	Laser Interferometer Space Antenna
LSST	Legacy Survey of Space and Time
NIR	Near infrared
NPIPE	Planck PR4 end-to-end processing pipeline
P/R/V	Principal, regular and vanishing-field homogeneous endpoints
ppm	Parts per million
PRIMAT	Precision BBN nuclear-rate compilation/code
PTA	Pulsar timing array
QR	Orthogonal-triangular matrix factorisation
Quaia	Gaia-unWISE quasar catalogue
S/V/T	Scalar/vector/tensor perturbation sectors
SDSS	Sloan Digital Sky Survey
SNR	Signal-to-noise ratio
SPARC	Spitzer Photometry and Accurate Rotation Curves database
SU(N)	Special unitary group of degree N
UV	Ultraviolet; high-energy completion
WISE	Wide-field Infrared Survey Explorer

Appendix C. Programmes and Reproducibility

The research code is predominantly Python and uses NumPy/SciPy for linear algebra and DOP853 integration, pandas for tabular outputs, Astropy/healpy for sky coordinates and masks, scikit-learn for propensity models, matplotlib for diagnostic figures and LINX/JAX for BBN.

Runs use preregistered case tables, SHA-256 manifests, deterministic seeds, per-case atomic checkpoints and restartable summaries.

The 19 August consistency audit parsed 56 scientific Python scripts with zero syntax failures, checked 270 manifest entries (265 local passes, no mismatches or missing files, five external identifiers) and independently rechecked 30/30 targeted SU(3) controls.

Table C1. Principal programme families and reproducibility artefacts.

Programme family	Representative output	Purpose
SU(2)/Quaia coherent null	su2_quaia_coherent_global_null_20260819/report.md	Catalogue-coherent global look-elsewhere calibration
SU(2) joint nuisance	su2_quaia_coherent_joint_nuisance_20260819	Refit scan-law/colour and quality model inside every mock
SU(3) identifiability	su3_identifiability_report_2026-08-17.md	Algebra, embeddings, invariant mapping and Jacobian rank
SU(3) basin/phase	su3_basin_map_*; su3_kappa_omega_*	Attractors, topology-loss and extinction boundaries
SU(3) finite-k	su3_finite_k_tensor_*; su3_full_scalar_closure_*; su3_lowxi_basin_finitek_svt_*	Constraint-reduced scalar/vector/tensor stability
SU(3) exit/reheating	su3_lowxi_dynamic_exit_*; su3_lowxi_qr_decay_transfer_*; su3_microscopic_reheating_*	Exit surfaces, orthonormal propagator and portal transfer
Consistency audit	su23_consolidated_consistency_audit_report_2026-08-19.md	Code, manifest, null and independent control review

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